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Identify and mitigate production issues by adopting AIOps in Azure Monitor

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Monitoring (the subtle art of having everything under control)

- Too many logs
- Difficulties in automating processes
- Difficulties in finding the issue





DevOps -> GitOps... and now AIOps

The DevOps methodology laid the foundations...

GitOps improved things... enhancing the world of infrastructure (and beyond...)

Now it's time for AIOps which improves IT, making it autonomous, predictive, and capable of self-optimization

Azure Monitor: the answer to (almost) all questions!

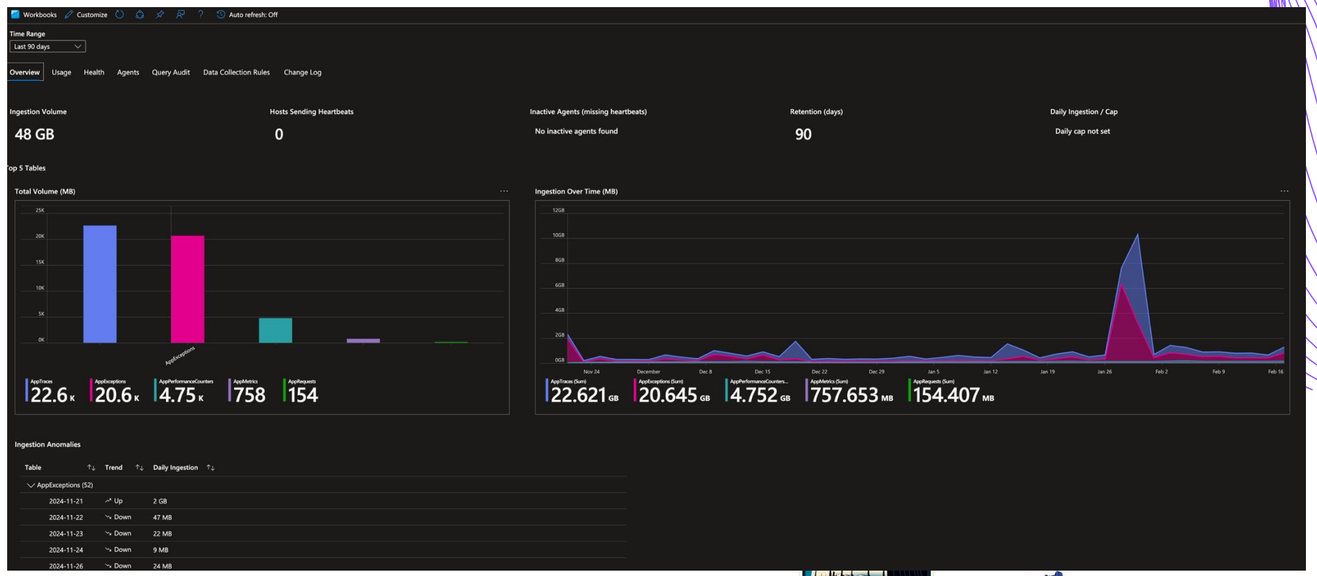
- Log Monitoring
- Application performance Monitoring
- Metric Alerts
- Virtual Machine Scale Set

Log Monitoring (Log Analytics Workspace Insights)

Azure Monitor provides a whole set of capabilities to analyze what's happening within the log workspace

In fact, it is possible to:

- monitor ingestion
- Finding anomalies
- Find the heaviest queries
- Check the operation of the agents on the machines
- Etc...



Log Monitoring (Kusto Query Language time series analysis and machine learning functions)

- Time series allow you to analyze any patterns in the logs through a time graph.
- Using the make-series command, you can easily obtain

```
| make-series Aggregation = Function(value)
[by Categoria]
on TimestampColumn
from Inizio to Fine
step Intervallo
```

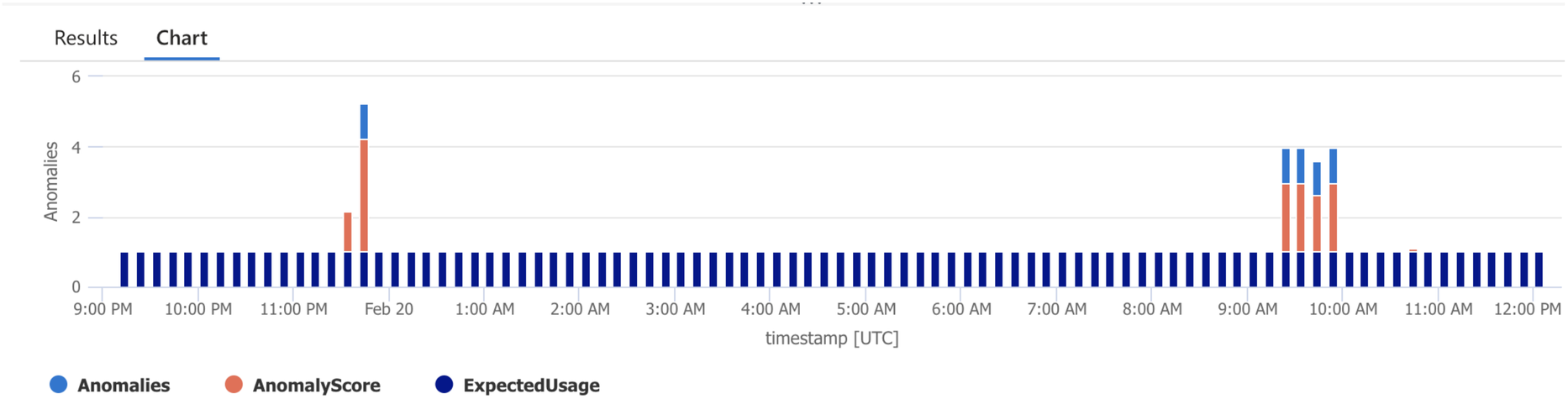
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Log Monitoring (Kusto Query Language time series analysis and machine learning functions)

- Once we have obtained a time series we can apply the function
- *series_decompose_anomalies(*Serie, Threshold [, Method]*)*
- The function analyzes the series and extrapolates the anomalies, giving it a score based on trend, and seasonality



Log Monitoring (Kusto Query Language time series analysis and machine learning functions)

Other features that implement ML

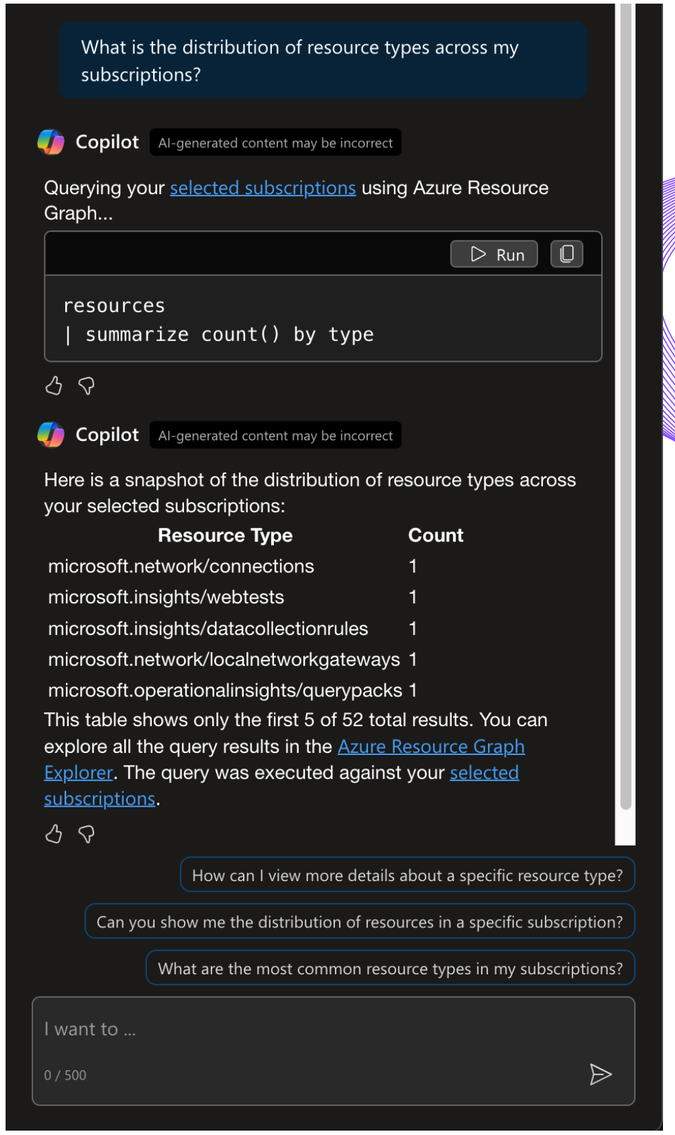
- **series_outliers():** Identification of anomalies in the data.
- **series_fit_line():** Linear regression on time series.
- **series_periods_detect():** Identify periodic pattern
- **series_fill_linear():** Linear interpolation for missing data.

Log Monitoring (Copilot in Azure)

Copilot for Azure helps us manage our log investigations using natural language

Use Case:

- Support for Kusto queries within it
- Search for information in subscriptions
- Have information on how much you are spending



Log Monitoring (Copilot in Azure)

There are limits though...

Any action performed on more than **10** resources must be performed outside of Microsoft Copilot in Azure.

You can only make **15** requests during a given chat, and you only have 10 chats in a **24-hour** period .

Some responses that display lists will be limited to the first **five** items.

For some tasks and queries, using a resource name won't work, and you'll need to provide the Azure resource ID.

Microsoft Copilot on Azure is currently only available in **English**.

DEMO TIME

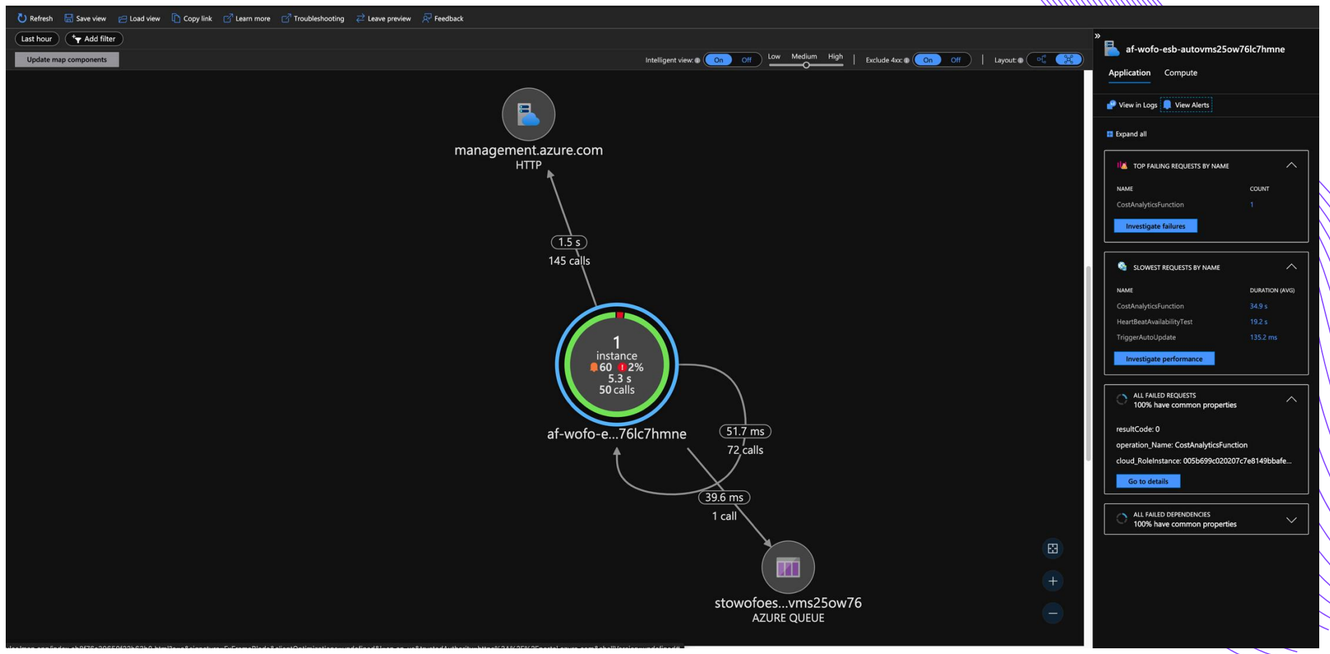


Application map intelligent mapping

It allows you to analyze distributed applications through the creation (via ML) of maps that highlight the connections between the various components

Components are the parts of the distributed application whose logs are visible to those who use the service.

It allows you to go into detail about the performance and errors encountered by each component, going all the way down to the call stack level

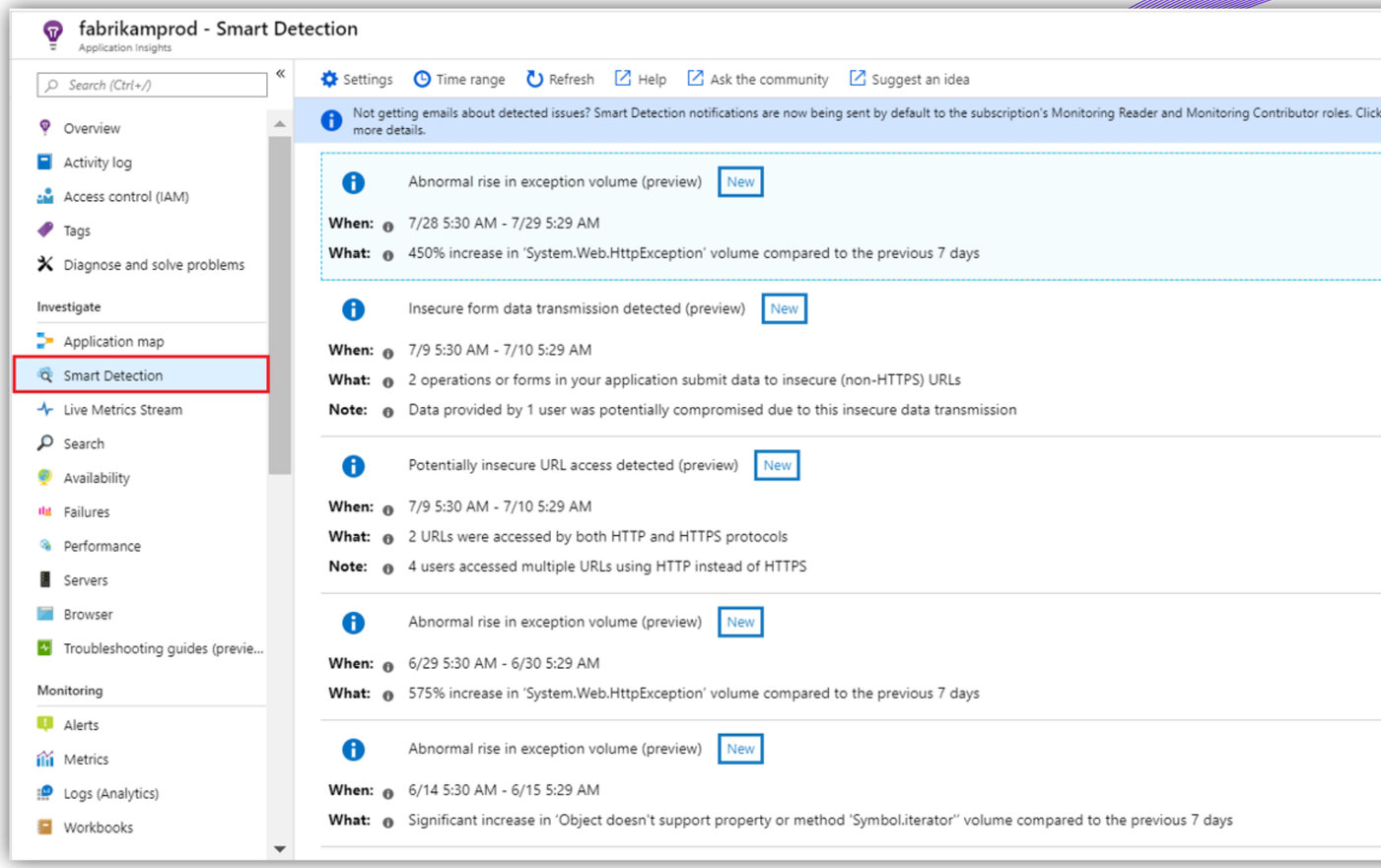


Smart Detection

Alerting + AI = Smart detection!

If the application sends enough information
, the AI will look for:

- Error and security patterns
- Abnormal mutations in the logs (many more than usual, for example...)
- Performance issues
- Memory leaks and much more...



Dynamic thresholds alerting

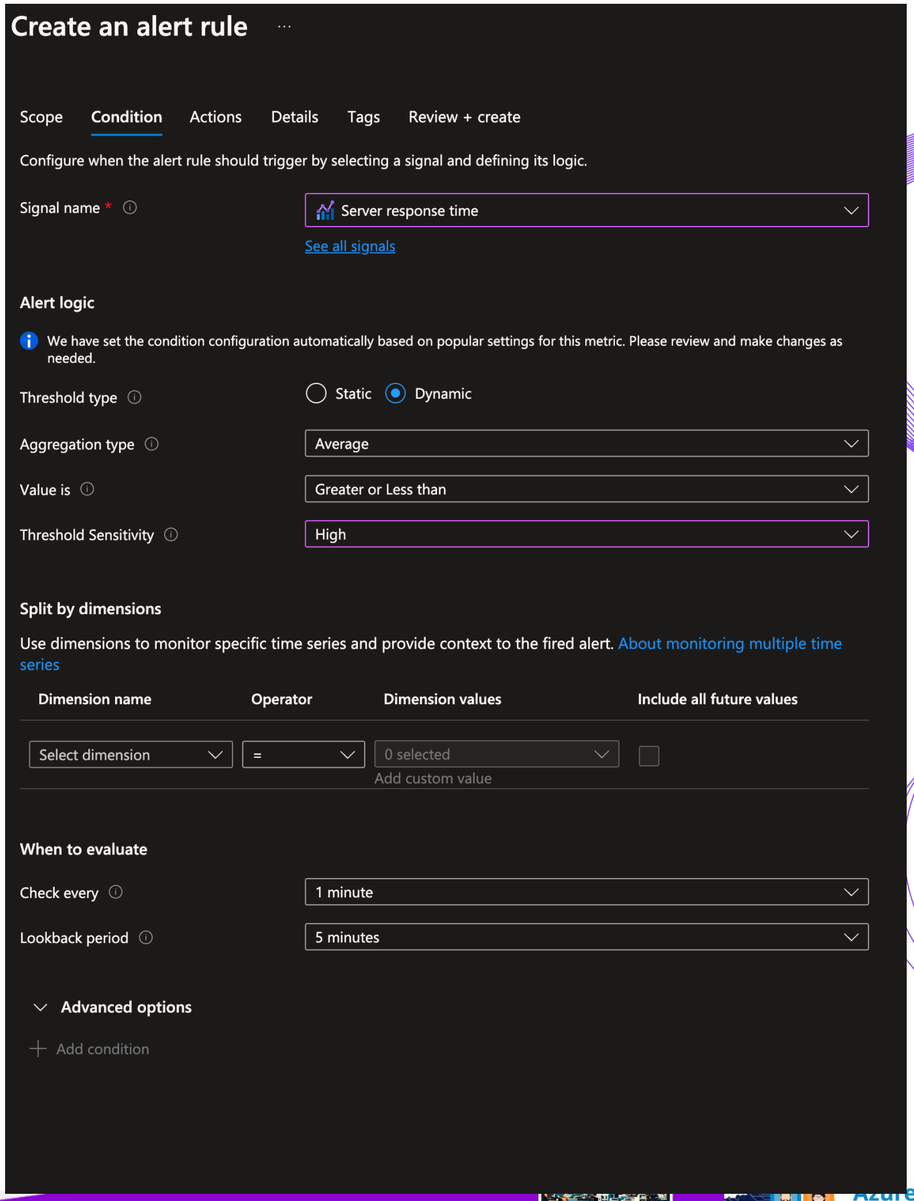
If you are unsure about which values to include in the alerts,

If you are not familiar with the behavior pattern of the resource,

If you don't want to create many alerts for so many resources to check the same thing (+ alert +€€),

You can use dynamic thresholds.

They use data from the last **10** days to "learn" and apply the most suitable thresholds

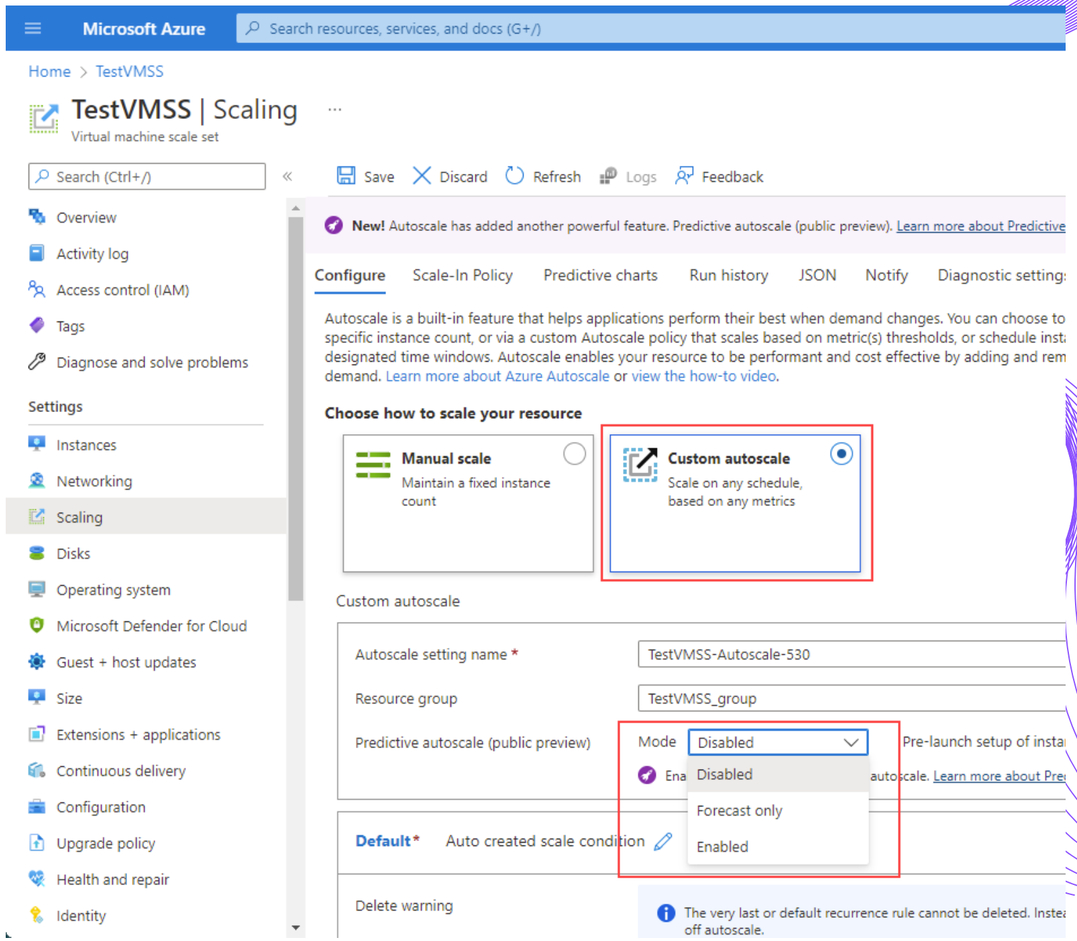


Predictive Autoscale

In VM scale sets that have a periodic load, it allows you to independently manage any load anomalies, before they occur

It uses historical data of about 15 days (rolling window) to "train" and requires at least 7 days of "training"

The "standard" autoscale is the reference in terms of when and how much to scale (as well as being a backup)



DEMO TIME



Bonus: ML e AIOps

Of course, with large amounts of data, KQL and Azure Monitor are no longer enough.

You can work with notebooks, or you can call the Azure Monitor APIs and then work the data in PySpark/Pandas DataFrame format

Or export the data directly to blob storage and start the pipeline from there

	Built-in KQL machine learning capabilities	Create your own machine learning pipeline
Scenario	✅ Anomaly detection, root cause, and time series analysis	✅ Anomaly detection, root cause, and time series analysis ✅ Advanced analysis and AIOPs scenarios
Advantages	🔹Gets you started very quickly. 🔹No data science knowledge and programming skills required. 🔹 Optimal performance and cost savings.	🔹Supports larger scales. 🔹Enables advanced, more complex scenarios. 🔹Flexibility in choosing libraries, models, parameters.
Service limits and data volumes	Azure portal or Query API log query limits depending on whether you're working in the portal or using the API, for example, from a notebook.	🔹Query API log query limits if you query data in Azure Monitor Logs as part of your machine learning pipeline. Otherwise, no Azure service limits. 🔹Can support larger data volumes.
Integration	None required. Run using Log Analytics in the Azure portal or from an integrated Jupyter Notebook.	Requires integration with a tool, such as Jupyter Notebook. Typically, you'd also integrate with other Azure services, like Azure Synapse Analytics.
Performance	Optimal performance, using the Azure Data Explorer platform, running at high scales in a distributed manner.	Introduces a small amount of latency when querying or exporting data, depending on how you implement your machine learning pipeline.
Model type	Linear regression model and other models supported by KQL time series functions with a set of configurable parameters.	Completely customizable machine learning model or anomaly detection method.
Cost	No extra cost.	Depending on how you implement your machine learning pipeline, you might incur charges for exporting data, ingesting scored data into Azure Monitor Logs, and the use of other Azure services.
Tutorial	Detect and analyze anomalies using KQL machine learning capabilities in Azure Monitor	Analyze data in Azure Monitor Logs using a notebook



Questions & Answers

Thank you for your attention!